

Презентация

Лабораторная работа №2

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Приобретение практических навыков по установке и конфигурированию DNS-сервера, усвоение принципов работы системы доменных имён.

Для начала запустим виртуальную машину через vagrant.

```
C:\work\dvkalashnikova\code\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
```

Рис. 1: Запуск ВМ

Теперь скачаем пакет bind utils.

```
=====
Package Architecture Version Repository Size
-----
Installing:
bind x86_64 32:9.18.33-15.el10_2.10 appstream 323 k
Installing weak dependencies:
bind-dnssec-utils x86_64 32:9.18.33-15.el10_2.10 appstream 147 k
Transaction Summary
-----
Install 2 Packages

Total download size: 470 k
Installed size: 1.3 M
Downloading Packages:
(1/2): bind-9.18.33-15.el10_2.10.x86_64.rpm 1.3 MB/s | 323 kB 00:00
(2/2): bind-dnssec-utils-9.18.33-15.el10_2.10.x86_64.rpm 552 kB/s | 147 kB 00:00
-----
Total 511 kB/s | 470 kB 00:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Installing : bind-dnssec-utils-32:9.18.33-15.el10_2.10.x86_64 1/2
  Running scriptlet: bind-32:9.18.33-15.el10_2.10.x86_64 2/2
  Installing : bind-32:9.18.33-15.el10_2.10.x86_64 2/2
  Running scriptlet: bind-32:9.18.33-15.el10_2.10.x86_64 2/2

Installed:
  bind-32:9.18.33-15.el10_2.10.x86_64 bind-dnssec-utils-32:9.18.33-15.el10_2.10.x86_64

Complete!
```

Рис. 2: Скачивание пакетов

Используем команду dig для проверки сервисов яндекса.

```
complete!
[root@server.dvkalashnikova.net ~]# dig www.yandex.ru

; <<> DiG 9.18.33 <<> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43034
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                187     IN      A      77.88.55.88
www.yandex.ru.                187     IN      A      77.88.44.55
www.yandex.ru.                187     IN      A      5.255.255.77

;; Query time: 29 msec
;; SERVER: 77.88.8.8#53(77.88.8.8) (UDP)
;; WHEN: Tue Sep 15 13:24:34 UTC 2026
;; MSG SIZE rcvd: 90
```

Рис. 3: dig ya.ru

Посмотрим на содержание файлов конфигурации dns в etc.

```
[root@server.dokshelkova.net ~]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
options {
    listen-on port 53 { 127.0.0.1 };
    listen-on-v6 port 53 { ::1 };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secrets-file "/var/named/data/named.secrets";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost };
}

/*
 * If you are building an AUTHORITY DNS server, do NOT enable recursion.
 * If you are building a RECURSIVE (caching) DNS server, you need to enable
 * recursion.
 * If your recursive DNS server has a public IP address, you MUST enable access
 * control to limit queries to your legitimate users. Failing to do so will
 * cause your server to become part of large scale DNS amplification
 * attacks. Implementing BCP38 within your network would greatly
 * reduce such attack surface
 */
recursion yes;

dnssec-validation yes;

named-key-directory "/var/named/dynamic";
geoip-directory "/usr/share/GeoIP";

pid-file "/run/named/named.pid";
session-keyfile "/run/named/session.key";

/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};
```

Рис. 4: Файлы конфигурации

Просмотрим теперь файл named.ca.

```
[root@server.dkalashnikov.net ~]# cat /var/named/named.ca
;
; This file holds the information on root name servers needed to
; initialize cache of Internet domain name servers
; (e.g. reference this file in the "cache . <file>"
; configuration file of BIND domain name servers).
;
; This file is made available by InterNIC
; under anonymous FTP as
;   file           /domain/named.cache
;   on server      FTP://INTERNIC.NET
;   -OR-           RS://INTERNIC.NET
;
; Last update:   December 20, 2023
; related version of root zone:  2023122001
;
; FORMERLY NS. INTERNIC.NET
;
;
; 30000000      NS      A.ROOT-SERVERS.NET.
A.ROOT-SERVERS.NET. 30000000      A      190.41.0.4
A.ROOT-SERVERS.NET. 30000000      AAAA   2801:503:ba3e::2:30
;
; FORMERLY NS1. ISI.EDU
;
; 30000000      NS      B.ROOT-SERVERS.NET.
B.ROOT-SERVERS.NET. 30000000      A      170.247.178.2
B.ROOT-SERVERS.NET. 30000000      AAAA   2801:1b8:10::b
;
; FORMERLY C. PSI.NET
;
; 30000000      NS      C.ROOT-SERVERS.NET.
C.ROOT-SERVERS.NET. 30000000      A      192.33.4.12
C.ROOT-SERVERS.NET. 30000000      AAAA   2801:500:2::c
;
; FORMERLY TEMP.UMD.EDU
;
; 30000000      NS      D.ROOT-SERVERS.NET.
D.ROOT-SERVERS.NET. 30000000      A      199.7.91.13
D.ROOT-SERVERS.NET. 30000000      AAAA   2801:500:2d::d
;
; FORMERLY NS.NASA.DOV
;
; 30000000      NS      E.ROOT-SERVERS.NET.
E.ROOT-SERVERS.NET. 30000000      A      192.203.230.10
E.ROOT-SERVERS.NET. 30000000      AAAA   2801:500:a0::e
;
; FORMERLY NS.ISC.ORG
;
; 30000000      NS      F.ROOT-SERVERS.NET.
F.ROOT-SERVERS.NET. 30000000      A      192.5.5.241
F.ROOT-SERVERS.NET. 30000000      AAAA   2801:500:2f::f
;
; FORMERLY NS.NIC.DEN.HIL
```

Рис. 5: named.ca

named.localhost и named.loopback

Содержимое named.localhost и named.loopback.

```

; End of file
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

NS     @
A      127.0.0.1
AAAA   ::1

[root@server.dvkalashnikova.net ~]# sudo cat /var/named/named.loopback
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
PTR     localhost.

[root@server.dvkalashnikova.net ~]#
```

Рис. 6: named.localhost и named.loopback

Запустим теперь named и осуществим снова dig yandex.ru.

```
vim /etc/named.conf
[root@server.dkcalashnikova.net ~]# systemctl start named
[root@server.dkcalashnikova.net ~]# systemctl enable named
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
[root@server.dkcalashnikova.net ~]# dig www.yandex.ru

;<<> DiG 9.18.33 <<> www.yandex.ru
;; global options: +cmd
;; Got answer:
;;->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39248
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 4096
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                388     IN      A      77.88.55.88
www.yandex.ru.                388     IN      A      5.255.255.77
www.yandex.ru.                388     IN      A      77.88.44.55

;; Query time: 27 msec
;; SERVER: 77.88.0.0[53] (77.88.0.0) (UDP)
;; WHEN: Tue Sep 15 13:29:51 UTC 2020
;; MSG SIZE rcvd: 90

[root@server.dkcalashnikova.net ~]# dig @127.0.0.1 www.yandex.ru
; communications error to 127.0.0.1[53]: timed out

;<<> DiG 9.18.33 <<> @127.0.0.1 www.yandex.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;;->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 8931
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: udp: 1232
; COOKIE: 3ab782018ed74061000000aa9485cef440531bffe5adf (good)
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; Query time: 778 msec
;; SERVER: 127.0.0.1[53] (127.0.0.1) (UDP)
;; WHEN: Tue Sep 15 13:30:04 UTC 2020
;; MSG SIZE rcvd: 70

[root@server.dkcalashnikova.net ~]#
```

Рис. 7: Запуск named

Теперь настроим порт eth0.

```
[root@server.dvkalashnikova.net ~]# nmcli connection edit eth0

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-lx, dcb, sriov, ethtool, match, ipv4, ipv6, prefix-delegation, hostname.
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (31ef6182-2c55-44b3-8944-a2af7d10d59a) successfully updated.
nmcli> quit
[root@server.dvkalashnikova.net ~]# nmcli connection edit System eth0
Error: Unknown connection 'System eth0'.
[root@server.dvkalashnikova.net ~]# systemctl restart NetworkManager
[root@server.dvkalashnikova.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search dvkalashnikova.net
nameserver 127.0.0.1
nameserver fd17:625c:0000:0000:0000:0000:0000:0000
[root@server.dvkalashnikova.net ~]#
```

Рис. 8: eth0

Откроем и отредактируем named.conf.

```
## DNS named 9.10
##
## named.conf
##
## Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
## server as a caching only nameserver (as a local host DNS resolver only).
##
## See /usr/share/doc/bind*/sample/ for example named configuration files.
##
options {
    listen-on port 53 { 127.0.0.1; any; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named.stats.txt";
    memstatistics-file "/var/named/data/named.mem.stats.txt";
    secrets-file "/var/named/data/named.secrets";
    recursive-file "/var/named/data/named.recursive";
    allow-query { localhost; 192.168.0.0/16; };

    /*
     * If you are building an AUTHORITY DNS server, do NOT enable recursion.
     * If you are building a RECURSIVE (caching) DNS server, you need to enable
     * recursion.
     * If your recursive DNS server has a public IP address, you MUST enable access
     * control to limit queries to your legitimate users. Failing to do so will
     * cause your server to become part of large scale DNS amplification
     * attacks. Implementing BCP38 within your network would greatly
     * reduce such attack surface
     */
    recursion yes;

    dnssec-validation yes;

    managed-keys-directory "/var/named/dynamic";
    group-directory "/usr/share/GeoIP";

    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";

    /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
    include "/etc/crypto-policies/back-ends/bind.conf";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};
```

Рис. 9: named.conf

Установим правила фаервола.

```
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=dns
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=dns --permanent
success
[root@server.dvkalashnikova.net ~]# ss -t | grep UDP
ss: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
ss: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
avahi-daemon 1120          avahi 12u      IPv4 0          0t0      *:*:dns
avahi-daemon 1120          avahi 13u      IPv6 0          0t0      *:*:dns
chronyd 1121             chrony 4u        IPv4 10111     0t0      localhost:323
chronyd 1121             chrony 5u        IPv6 10112     0t0      localhost:323
chronyd 1121             chrony 8u        sock 0          0t0      41978 protocol: UDP
wsdd 13585              dvkalashnikova 7u      IPv4 42290     0t0      UDP 239.255.255.250:ws-discovery
wsdd 13585              dvkalashnikova 8u      IPv4 42291     0t0      UDP *:39453
wsdd 13585              dvkalashnikova 9u      IPv4 42292     0t0      UDP server.dvkalashnikova.net:ws-discovery
```

Рис. 10: Правила фаервола

Теперь переместим файл с настройкой конфига.

```
[root@server.dvkalashnikova.net ~]# cp /etc/named.rfc1912.zones /etc/named/  
[root@server.dvkalashnikova.net ~]# cd /etc/named  
[root@server.dvkalashnikova.net named]# mv /etc/named/named.rfc1912.zones /etc/named/dv  
[root@server.dvkalashnikova.net named]# nano /etc/named.conf  
[root@server.dvkalashnikova.net named]# nano /etc/named/user.net  
[root@server.dvkalashnikova.net named]# nano /etc/named/dvkalashnikova.net
```

Рис. 11: перемещение файла

И отредактируем наш файл под наши параметры.

```
bind named 0.1
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localst DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
options {
    listen-on port 53 { 127.0.0.1; any; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named.stats.txt";
    memstatistics-file "/var/named/data/named.mem.stats.txt";
    secrets-file "/var/named/data/named.secrets";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; 192.168.0.0/16; };
}

/*
- If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
- If you are building a RECURSIVE (caching) DNS server, you need to enable
  recursion.
- If your recursive DNS server has a public IP address, you MUST enable access
  control to limit queries to your legitimate users. Failing to do so will
  cause your server to become part of large scale DNS amplification
  attacks. Implementing BCP38 within your network would greatly
  reduce such attack surface
*/
recursion yes;

dnssec-validation yes;

managed-keys-directory "/var/named/dynamic";
geoip-directory "/usr/share/GeoIP";

pid-file "/run/named/named.pid";
session-keyfile "/run/named/session.key";

/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
include "/etc/named/dynamic/keys/named.key";
```

То же самое сделаем с файлом зон.

```
GNU nano 8.1
// named.rfc1912.zones:
//
// Provided by Red Hat caching-nameserver package
//
// ISC BIND named zone configuration for zones recommended by
// RFC 1912 section 4.1 : localhost TLDs and address zones
// and https://tools.ietf.org/html/rfc5303
// (c)2007 R W Franks
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add
// disable-empty-zone "."; into options
//
zone "dvkalashnikova.net" IN {
    type master;
    file "master/fz/dvkalashnikova.net";
    allow-update { none; };
};

zone "1.168.192.in-addr.arpa" IN {
    type master;
    file "master/rz/192.168.1";
    allow-update { none; };
};
```

Рис. 13: Файл зон

Создадим папки с настройками днс.

```
[root@server.dvkalashnikova.net named]# cd /var/named
[root@server.dvkalashnikova.net named]# mkdir -p /var/named/master/fz
[root@server.dvkalashnikova.net named]# mkdir -p /var/named/master/tz
[root@server.dvkalashnikova.net named]# cp /var/named/named.localhost /var/named/master/fz/
[root@server.dvkalashnikova.net named]# cd /var/named/master/fz/
[root@server.dvkalashnikova.net fz]# mv named.localhost dvkalashnikova.net
[root@server.dvkalashnikova.net fz]# nano /var/named/master/fz/dvkalashnikova.net
```

Рис. 14: Создание папок и настроек днс

Отредактируем файл dvkalashnikova.net.

```
GNU nano 8.1
$TTL 1D
@      IN SOA  @ server.dvkalashnikova.net. (
                                @      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

      NS      @
      A       192.168.1.1
      AAAA    ::1
$ORIGIN dvkalashnikova.net.
server A 192.168.1.1
ns A 192.168.1.1
```

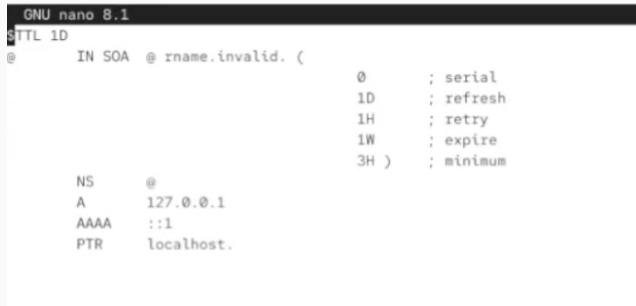
Рис. 15: dvkalashnikova.net

Теперь посмотрим на файлы из папки rz.

```
[root@server.dvkalashnikova.net fz]# cp /var/named/named.loopback /var/named/master/rz/  
[root@server.dvkalashnikova.net fz]# cd /var/named/master/rz/  
[root@server.dvkalashnikova.net rz]# mv named.loopback 192.168.1  
[root@server.dvkalashnikova.net rz]# nano
```

Рис. 16: Папка rz

Отредактируем следующим образом.



```
GNU nano 8.1
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )    ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
PTR    localhost.
```

Рис. 17: Редактирование файла

Настроим Selinux.

```
[root@server.dvkalashnikova.net rz]# chown -R named:named /etc/named
[root@server.dvkalashnikova.net rz]# chown -R named:named /var/named
[root@server.dvkalashnikova.net rz]# restorecon -vR /etc
[root@server.dvkalashnikova.net rz]# restorecon -vR /var/named
[root@server.dvkalashnikova.net rz]# getsebool -a | grep named
named_tcp_bind_http_port --> off
named_write_master_zones --> on
[root@server.dvkalashnikova.net rz]# setsebool named_write_master_zones 1
[root@server.dvkalashnikova.net rz]# setsebool -P named_write_master_zones 1
[root@server.dvkalashnikova.net rz]#
```

Рис. 18: Selinux

Через dig попробуем подключиться к собственному днс.

```
[root@server.dkcalashnikova.net rz]# dig ns.dkcalashnikova.net

;<<> DIG 9.18.33 <<> ns.dkcalashnikova.net
;; global options: +cmd
;; Out answer:
;;>>HEADER<<< opcode: QUERY, status: NOERROR, id: 14708
;; Flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 1232
;; COOKIE: 0a0b701446a0857818000000ea9533be773cfd604e7612 (good)
;; QUESTION SECTION:
;ns.dkcalashnikova.net.      IN      A
;; ANSWER SECTION:
ns.dkcalashnikova.net.      86400   IN      A      192.168.1.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Tue Sep 15 14:16:27 UTC 2009
;; MSG SIZE  rcvd: 94

[root@server.dkcalashnikova.net rz]# host -l dkcalashnikova.net
dkcalashnikova.net name server dkcalashnikova.net.
dkcalashnikova.net has address 192.168.1.1
dkcalashnikova.net has IPv6 address ::1
ns.dkcalashnikova.net has address 192.168.1.1
server.dkcalashnikova.net has address 192.168.1.1
[root@server.dkcalashnikova.net rz]# host -a dkcalashnikova.net
Trying "dkcalashnikova.net"
;<<>HEADER<<< opcode: QUERY, status: NOERROR, id: 59508
;; Flags: qr aa rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;dkcalashnikova.net.      IN      ANY
;; ANSWER SECTION:
dkcalashnikova.net.      86400   IN      SOA      dkcalashnikova.net. server.dkcalashnikova.net. 0 86400 3600 604800 10000
dkcalashnikova.net.      86400   IN      NS      dkcalashnikova.net.
dkcalashnikova.net.      86400   IN      A      192.168.1.1
dkcalashnikova.net.      86400   IN      AAAA     ::1

Received 137 bytes from 127.0.0.1#53 in 0 ms
[root@server.dkcalashnikova.net rz]# host -t A dkcalashnikova.net
dkcalashnikova.net has address 192.168.1.1
[root@server.dkcalashnikova.net rz]# host -t PTR 192.168.1.1
1.1.168.192.in-addr.arpa domain name pointer ns.dkcalashnikova.net.
1.1.168.192.in-addr.arpa domain name pointer server.dkcalashnikova.net.
[root@server.dkcalashnikova.net rz]#
```

Рис. 19: dig

Оформим нашу работу как конфигурацию для вагранта.

```
[root@server.dvkalashnikova.net rz]# cd /vagrant
[root@server.dvkalashnikova.net vagrant]# mkdir -p /vagrant/provision/server/dns/etc/named
[root@server.dvkalashnikova.net vagrant]# mkdir -p /vagrant/provision/server/dns/var/named/master/
[root@server.dvkalashnikova.net vagrant]# cp -R /etc/named.conf /vagrant/provision/server/dns/etc/
[root@server.dvkalashnikova.net vagrant]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
[root@server.dvkalashnikova.net vagrant]# cp -R /var/named/master/* /vagrant/provision/server/dns/var/named/mast
[root@server.dvkalashnikova.net vagrant]# touch dns.sh
[root@server.dvkalashnikova.net vagrant]# chmod +x dns.sh
[root@server.dvkalashnikova.net vagrant]# nano dns.sh
```

Рис. 20: Конфиг вагрант

И напишем скрипт для загрузки вагранта.

```
GNU nano 0.1
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install bind bind-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/dns/etc/* /etc
cp -R /vagrant/provision/server/dns/var/named/* /var/named
chown -R named:named /etc/named
chown -R named:named /var/named
restorecon -vR /etc
restorecon -vR /var/named
echo "Configure firewall"
firewall-cmd --add-service=dns
firewall-cmd --add-service=dns --permanent
echo "Tuning SELinux"
setsebool named_write_master_zones 1
setsebool -P named_write_master_zones 1
echo "Change dns server address"
nmcli connection edit "System eth0" <<EOF
remove ipv4.dns
set ipv4.ignore-auto-dns yes
set ipv4.dns 127.0.0.1
save
quit
EOF
systemctl restart NetworkManager
echo "Start named service"
systemctl enable named
systemctl start named
```

Рис. 21: скрипт

И в vagrantfile будем загружать этот скрипт.

```
server.vm.provision "server dummy", type: "shell", preserve_order: true, path: "provision/server/01-dummy.sh"
server.vm.provision "server forward", type: "shell", preserve_order: true, run: "always", path: "provision/server/02-forward.sh"
server.vm.provision "server dns", type: "shell", preserve_order: true, path: "provision/server/dns.sh"
end
```

Рис. 22: vagrantfile

В результате выполнения работы были получены навыки настройки днс.